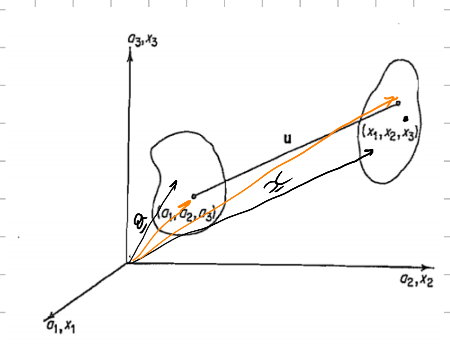
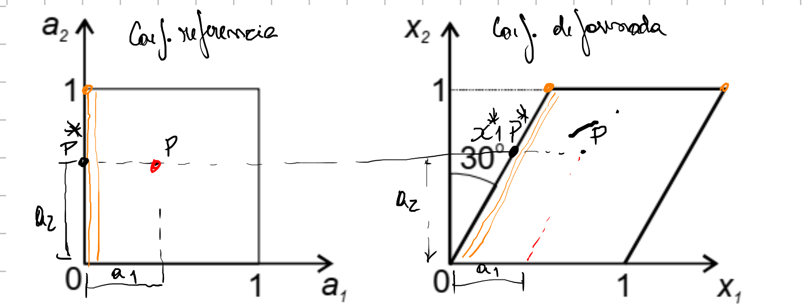


$$\underline{x} = \underline{x}^*(\overbrace{a_1, a_2, a_3, t}^{\text{uma partícula}})$$

$$\underline{a} = \underline{a}^*(x_1, x_2, x_3, t)$$

$$\underline{u} = \underline{x} - \underline{a}$$



$$E_{ij} = \frac{1}{2}\left(\delta_{\alpha\beta}\frac{\partial x_\alpha}{\partial a_i}\frac{\partial x_\beta}{\partial a_j} - \delta_{ij}\right),$$

$$x_1 = \overbrace{a_2 \cdot tg(30^\circ)}^{x_1^*} + a_1 \qquad x_3 = a_3$$

$$x_2 = a_2$$

$$\underline{E_{ij}} = \frac{1}{2}\left(\delta_{\alpha\beta} \cdot \frac{\partial x_\alpha}{\partial a_i}\frac{\partial x_\beta}{\partial a_j} - \delta_{ij}\right) = \frac{1}{2}\left(\frac{\partial x_\alpha}{\partial a_i} \cdot \frac{\partial x_\alpha}{\partial a_j} - \delta_{ij}\right)$$

$$E_{11} = \frac{1}{2}\left(\frac{\partial x_1}{\partial a_1} \cdot \frac{\partial x_1}{a_1} + \frac{\partial x_2}{\partial a_1} \cdot \frac{\partial x_2}{\partial a_1} + \frac{\partial x_3}{\partial a_1} \cdot \frac{\partial x_3}{\partial a_1} - \overset{\delta_{11}}{1}\right)$$

$$= \frac{1}{2}\left(1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 - 1\right) = 0$$

$$E_{22} = \frac{1}{2}\left(\overset{tg 30^\circ}{\frac{\partial x_1}{\partial a_2}} \cdot \overset{tg 30^\circ}{\frac{\partial x_1}{\partial a_2}} + \overset{1}{\frac{\partial x_2}{\partial a_2}} \cdot \overset{1}{\frac{\partial x_2}{\partial a_2}} + \overset{0}{\frac{\partial x_3}{\partial a_2}} \cdot \overset{0}{\frac{\partial x_3}{\partial a_2}} - 1\right)$$

$$= \frac{1}{2}\left(tg^2 30^\circ\right)$$

$$E_{33} = \frac{1}{2}\left(1 - 1\right) = 0$$

$$E_{12} = \frac{1}{2}\left(\overset{1}{\frac{\partial x_1}{\partial a_1}} \cdot \overset{tg(30^\circ)}{\frac{\partial x_1}{\partial a_2}} + \overset{0}{\frac{\partial x_2}{\partial a_1}} \cdot \overset{1}{\frac{\partial x_2}{\partial a_2}} + \overset{0}{\frac{\partial x_3}{\partial a_1}} \cdot \overset{0}{\frac{\partial x_3}{\partial a_2}} - 0\right) = \frac{1}{2}(tg 30^\circ)$$

$$E_{21} = E_{12} = 1/2\, tg(30^\circ)$$

$$E_{31} = \frac{1}{2}\left(\overset{0}{\cancel{\frac{\partial x_1}{\partial a_3}}} \cdot \overset{1}{\cancel{\frac{\partial x_1}{\partial a_1}}} + \overset{0}{\cancel{\frac{\partial x_2}{\partial a_3}}} \cdot \overset{0}{\cancel{\frac{\partial x_2}{\partial a_1}}} + \overset{1}{\cancel{\frac{\partial x_3}{\partial a_3}}} \cdot \overset{0}{\cancel{\frac{\partial x_3}{\partial a_1}}} - 0\right) = 0 = E_{13}$$

$E_{32} = 0 = E_{23}$

$tg(30°) = \frac{\sqrt{3}}{3} \Rightarrow tg^2(30°) = \frac{3}{9} = \frac{1}{3}$

$$\underline{\underline{E}} = \begin{bmatrix} 0 & 1/2\, tg(30°) & 0 \\ & 1/2\, tg^2(30°) & 0 \\ \text{sym} & & 0 \end{bmatrix} = \begin{bmatrix} 0 & \sqrt{3}/6 & 0 \\ & 1/6 & 0 \\ & & 0 \end{bmatrix}$$

$$e_{ij} = \frac{1}{2}\left(\delta_{ij} - \delta_{\alpha\beta}\frac{\partial a_\alpha}{\partial x_i}\frac{\partial a_\beta}{\partial x_j}\right),$$

$a_\alpha(x_1, x_2, x_3)$

$$e_{ij} = \frac{1}{2}\left(\delta_{ij} - \frac{\partial a_\alpha}{\partial x_i} \cdot \frac{\partial a_\alpha}{\partial x_j}\right)$$

$x_1 = a_2 . tg(30°) + a_1$

$x_3 = a_3 \Rightarrow a_3 = x_3$

$x_2 = a_2 \Rightarrow a_2 = x_2$

$x_1 = x_2 . tg(30°) + a_1 \Rightarrow a_1 = x_1 - x_2\, tg(30°)$

$$\begin{cases} a_1 = x_1 - x_2\, tg(30°) \\ a_2 = x_2 \\ a_3 = x_3 \end{cases}$$